CIDA: Collective Intelligence via Deliberation and Aggregation

Kairat Zhaksylykov
Student, K.Zhubanov Regional University
zhaksylykov.k06@gmail.com

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Abstract

Modern neural networks, even when pre-trained, often suffer from miscalibration: their confidence scores do not align with actual correctness. We introduce CIDA (Collective Intelligence via Deliberation and Aggregation), a text classification architecture where N agents engage in an iterative deliberation protocol (CDP) — exchanging compressed arguments (theses), updating their beliefs through a recurrent mechanism, and reaching a consensus weighted by each agent's uncertainty. When integrated with BERT-tiny (CIDA-BERT), our model reduces Expected Calibration Error (ECE) by 38% (from 9.09% to 5.63%) on SST-2 while preserving accuracy (83.14% vs 83.60%). Furthermore, a CIDA model trained from scratch with only a simple token embedder achieves 76.7% accuracy and 3.4% ECE, compared to 50.9% and 5.7% for the baseline without deliberation — proving that CDP's benefits stem from its architecture, not from additional parameters. Our ablation study isolates the contribution of each component: sparse token communication is critical for calibration, while the number of agents and loss terms provide complementary regularization. All code and results are reproducible.

1 Introduction

State-of-the-art text classifiers achieve high accuracy but are frequently miscalibrated: they assign overconfident probabilities, especially on out-of-distribution inputs or ambiguous examples. For instance, BERT-tiny yields an Expected Calibration Error (ECE) of 9.09% on SST-2, meaning its confidence is systematically too high. This is problematic for high-stakes applications such as medical diagnosis, legal analysis, and autonomous systems.

Existing calibration techniques include post-processing methods (temperature scaling [2]) and ensembles [6]. However, temperature scaling requires a hold-out set and does not change the model's representations, while ensembles multiply inference cost.

We propose a fundamentally different approach: embed a multi-agent deliberation mechanism directly into the architecture. Inspired by multi-agent debate [5] and the Slot Attention mechanism [3], our Collective Deliberation Protocol (CDP) lets N agents form initial perspectives, exchange compact arguments (theses) over several rounds, update their beliefs via a gated recurrent cell, and finally aggregate their outputs with weights inversely proportional to their uncertainty.

Our main contributions:

- We present CIDA V8, the first architecture that uses Slot Attention to generate agent beliefs for text classification, combined with Sparse Token Communication, Agent-LoRA, and a local Sparse Mixture of Experts inside each deliberation round.
- We show that CDP improves calibration significantly: when plugged into BERT-tiny, ECE drops by 38% (9.09% $\rightarrow$ 5.63%) while accuracy remains virtually unchanged (83.60% $\rightarrow$ 83.14%).
- Through an ablation study on a simple token embedder (without pre-training), we prove that CDP is not merely adding parameters: the full CDP raises accuracy from 50.9% (random guessing) to 76.7% with only a 20% parameter increase, whereas simply widening the baseline yields no improvement.
- We analyse the contribution of each component (communication, diversity loss, dominance loss, number of agents) and provide scaling laws for limited-data regimes.

2 Related Work

2.1 Calibration of Neural Networks

Guo et al. [2] systematically studied miscalibration in deep networks and proposed temperature scaling as a simple post-hoc fix. While effective, it does not alter the model's internal representations. Ensemble methods [6] improve calibration but multiply inference cost. In contrast, CDP makes calibration endogenous through architectural design.

2.2 Multi-Head Attention and Transformers

The Transformer [1] aggregates information from multiple heads via concatenation and projection. Heads operate independently; there is no direct communication in the representation space. CDP introduces explicit inter-agent communication.

2.3 Slot Attention

Slot Attention [3] was originally designed for object-centric learning in vision. It enforces competition among slots by normalising attention over slots rather than over positions. We adapt it to NLP: slots become agents that compete for semantic aspects of the input text.

2.4 Mixture of Experts

Sparse Mixture of Experts (MoE) architectures [4, 7] increase model capacity at constant computation cost. In CIDA, a local SparseMoE is used inside each deliberation round, allowing agents to specialise in different reasoning patterns.

2.5 Multi-Agent Debate

Du et al. [5] showed that iterative debates between LLMs improve factual accuracy. Unlike their post-hoc approach, CIDA trains the entire debate process end-to-end with compact latent theses, enabling backpropagation through the deliberation steps.

3 Method: CIDA V8 Architecture

CIDA V8 consists of six sequential modules (Fig. **??**). First, input text is tokenised with Byte-Pair Encoding (vocab size 16k). A Transformer encoder (or optionally a simple embedder) produces contextualised token representations. The Slot Attention Perspective Generator creates N initial agent beliefs through a competitive attention mechanism. Then, R rounds of the `DeliberationLayer` let agents exchange theses and update their states. After deliberation, Uncertainty Heads estimate each agent's predictive uncertainty, and a Consensus Aggregation module produces a weighted consensus. Finally, a MetaReviewer outputs class logits. A residual connection from the encoder's CLS token provides a skip connection.

4 Theoretical Analysis

We provide a mathematical justification for why the Collective Deliberation Protocol (CDP) improves calibration and leads to stable consensus.

4.1 Notation and Preliminaries

Let $\mathbf{x} \in \mathcal{X}$ be an input, $y \in \{1, \ldots, C\}$ the true label. For a model f producing probabilities $\hat{p}(y|\mathbf{x})$, the *Expected Calibration Error* (ECE) is defined as

$$\mathrm{ECE} = \mathbb{E}_{\hat{p}}\left[|\Pr(y = \hat{y} \mid \hat{p} = p) - p|\right],$$

which measures the deviation between predicted confidence and actual accuracy.

In CDP, we have N agents. Each agent i outputs a feature vector $\mathbf{f}_i(\mathbf{x})$ and a scalar variance $v_i(\mathbf{x})$. The consensus prediction is

$$\mathbf{F}(\mathbf{x}) = \sum_{i=1}^{N} w_i(\mathbf{x})\mathbf{f}_i(\mathbf{x}), \quad w_i(\mathbf{x}) = \frac{\exp(-\log v_i(\mathbf{x}) + \log \tau_i)}{\sum_j \exp(-\log v_j(\mathbf{x}) + \log \tau_j)},$$

where τ_i are learnable trust parameters. This weighting is a softmax over the log of inverse variances, i.e. $w_i \propto 1/v_i$ times a prior trust.

4.2 Optimal Weighting Minimises Consensus Variance

Assume each agent's prediction $\mathbf{f}_i$ is an unbiased estimate of some true latent representation $\boldsymbol{\mu}(\mathbf{x})$, with variance $\mathrm{Var}(\mathbf{f}_i) = v_i(\mathbf{x})$. For simplicity, consider scalar outputs f_i. The consensus is $F = \sum_i w_i f_i$ with $\sum_i w_i = 1$. The variance of the consensus is

$$\mathrm{Var}(F) = \sum_i w_i^2 v_i + 2\sum_{i<j} w_i w_j \mathrm{Cov}(f_i, f_j).$$

If the agents are conditionally independent given $\mathbf{x}$ (a reasonable assumption due to different specialisations and the decorrelation loss L_{div}), then $\mathrm{Cov}(f_i, f_j) \approx 0$. Minimising $\sum_i w_i^2 v_i$ subject to $\sum_i w_i = 1$ yields

$$w_i^* = \frac{1/v_i}{\sum_j 1/v_j},$$

which is exactly the inverse-variance weighting (homoscedastic case). Therefore, the CDP consensus achieves the minimum possible variance among all linear unbiased estimators. Lower variance directly leads to better calibration, because overconfidence is often caused by high variance in logits.

4.3 Convergence of Iterative Deliberation

Each deliberation round updates beliefs via a GRU cell:

$$\mathbf{b}^{(t)} = \mathrm{GRU}\left(\mathbf{a}^{(t)}, \mathbf{b}^{(t-1)}\right),$$

where $\mathbf{a}^{(t)}$ is the aggregated information from other agents (theses). We model this as a dynamical system:

$$\mathbf{b}^{(t)} = \mathcal{T}(\mathbf{b}^{(t-1)}, \mathbf{x}),$$

where $\mathcal{T}$ is a Lipschitz continuous operator. The loss component $L_{\text{conv}} = \frac{1}{R-2}\sum_{t\geq 2}\|\mathbf{b}^{(t)} - \mathbf{b}^{(t-1)}\|$ encourages the sequence to converge to a fixed point $\mathbf{b}^*$.

Proposition 1 (Contraction). If the GRU cell satisfies $\|\frac{\partial \mathcal{T}}{\partial \mathbf{b}}\| \leq \gamma < 1$ uniformly (e.g., due to the gating mechanism), then the deliberation process is a contraction mapping. Consequently, the sequence $\{\mathbf{b}^{(t)}\}$ converges exponentially fast to a unique fixed point $\mathbf{b}^*$, and the convergence rate is $O(\gamma^t)$.

The gating update in the deliberation layer further stabilises the iteration:

$$\mathbf{b}_{\text{new}} = \mathbf{b}_{\text{old}} + \mathbf{g} \odot \mathbf{u}, \quad \mathbf{g} \in (0,1)^d,$$

which is a convex combination of old and new states, guaranteeing boundedness.

4.4 Variance Reduction via Multi-Agent Averaging

Consider the final consensus logits $z = \sum_i w_i z_i$. For a classification task, the softmax probabilities are $p_c = e^{z_c}/\sum_k e^{z_k}$. Assuming the logits z_i are i.i.d. around the true logits μ with covariance Σ_i, the variance of p_c can be approximated via the delta method:

$$\text{Var}(p_c) \approx \left(\frac{\partial p_c}{\partial \mathbf{z}}\right)^\top \left(\sum_i w_i^2 \Sigma_i\right) \left(\frac{\partial p_c}{\partial \mathbf{z}}\right).$$

Since $\sum_i w_i^2 \leq 1$ (with equality only for $N = 1$), the variance is reduced by a factor of up to $1/N$ compared to a single agent. This explains why CDP produces less overconfident (better calibrated) predictions: the averaging of several slightly decorrelated agents smooths the logits, reducing extreme values.

4.5 Connection to Calibration

A well-known fact in ensemble learning is that the average of unbiased predictors with uncorrelated errors has lower mean squared error (MSE). For calibration, we care about the Brier score or negative log-likelihood. Under the assumption that the agents' predictive distributions are drawn from a distribution centred at the true distribution, the consensus distribution (geometric or arithmetic mean) is closer to the truth in terms of KL divergence. Specifically, if each agent outputs a distribution $q_i(y|\mathbf{x})$, the consensus $q_{\text{cdp}} = \sum_i w_i q_i$ minimises the expected KL divergence to the true distribution p over the simplex of weights. This is a direct consequence of the convexity of KL divergence: the barycenter of distributions improves calibration.

Moreover, the explicit uncertainty head v_i learns to predict the agent's own variance. When the agent is uncertain (high v_i), its weight w_i decreases, preventing overconfident mistakes. This mechanism is analogous to a Bayesian model averaging with variance estimates.

4.6 Summary of Theoretical Guarantees

- Inverse-variance weighting gives the minimum variance linear unbiased estimator.
- The GRU-based deliberation is a contraction under mild Lipschitz conditions, guaranteeing convergence to a fixed point.
- Multi-agent averaging reduces predictive variance by a factor up to $1/N$, which directly improves calibration metrics (ECE, Brier score).
- The diversity loss L_{div} enforces low correlation among agents, making the variance reduction near-optimal.

These theoretical results are empirically validated in our ablation study (Section **??**) and main experiments.

4.7 Slot Attention Perspective Generator

We adopt Slot Attention with N slots (agents). For each of four iterations:

$$\mathbf{Q} = \text{proj}_q(\text{norm}(\text{slots})), \quad \mathbf{K} = \text{proj}_k(\text{norm}(\text{input})), \quad \mathbf{V} = \text{proj}_v \tag{1}$$

$$\text{attn} = \text{softmax}_{\text{slots}}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d}) \tag{2}$$

$$\text{updates} = \text{norm}(\text{attn}^\top \mathbf{V})/(\sum \text{attn}) \tag{3}$$

$$\text{slots} = \text{GRU}(\text{SwiGLU}(\text{updates}), \text{slots_prev}) \tag{4}$$

Normalisation over slots enforces competition: if one agent attends strongly to a token, others have less access, encouraging specialisation.

4.8 Deliberation Layer

Each deliberation round consists of three steps (Fig. **??**):

Sparse Token Communication (Theses). Each agent compresses its belief into $K = 4$ theses using a linear bottleneck: $\mathbf{theses}_i = \text{Linear}(d \to 4d)(\mathbf{belief}_i)$. All theses are concatenated, and each agent attends to the joint thesis set: $\mathbf{attended}_i = \text{CrossAttn}(\mathbf{belief}_i, \mathbf{theses}_{\text{all}}, \mathbf{theses}_{\text{all}})$.

AgentLoRA. Each agent applies a low-rank adaptation (rank $r = 8$) to personalise its interpretation: $\mathbf{out}_i = \mathbf{x} + \mathbf{x}\mathbf{A}_i\mathbf{B}_i$, where $\mathbf{A}_i \in \mathbb{R}^{d\times r}$, $\mathbf{B}_i \in \mathbb{R}^{r\times d}$.

Sparse MoE. Each agent is routed to top-2 of 4 experts via a linear gate: $\mathbf{expert_out} = \sum_{\text{top-2}} \text{softmax}(g) \cdot \text{SwiGLU}_{\text{expert}}(\mathbf{x})$.

Finally, a gated update mixes the old belief with the processed representation: $\mathbf{gate} = \sigma(\mathbf{W}_{\text{gate}}[\mathbf{h}_{\text{old}}; \mathbf{h}_{\text{new}}])$, $\mathbf{h}_{\text{new}} = \mathbf{h}_{\text{old}} + \mathbf{gate} \cdot \text{dropout}(\mathbf{h}_{\text{new}})$. The same layer is applied repeatedly across rounds (weight tying), turning deliberation into an iterative algorithm.

4.9 Uncertainty Heads and Consensus Aggregation

After the final round, each agent i outputs a feature vector and a scalar variance: $\mathbf{feat}_i = \mathrm{Linear}(\mathbf{b}_i)$, $\mathrm{var}_i = \mathrm{softplus}(\mathrm{Linear}_{\text{var}}(\mathbf{b}_i)) + \epsilon$.

Consensus weights are a product of inverse uncertainty and a learnable trust parameter:

$$w_{\text{unc}} = \mathrm{softmax}(-\mathrm{var}/T), \quad w_{\text{trust}} = \mathrm{softmax}(\mathbf{trust}), \tag{5}$$

$$w_{\text{final}} = \mathrm{softmax}(\log w_{\text{unc}} + \log w_{\text{trust}}), \quad \mathbf{c} = \sum_i w_{\text{final},i} \cdot \mathbf{feat}_i, \tag{6}$$

with temperature $T = \exp(\theta_{\text{temp}})/R_{\text{final}}$.

4.10 Loss Function

The total loss has six components:

$$L = L_{\text{task}} + \lambda_{\text{conv}} L_{\text{conv}} + \lambda_{\text{nt}} L_{\text{nt}} + \lambda_{\text{div}} L_{\text{div}} + \lambda_{\text{dom}} L_{\text{dom}} + \lambda_{\text{ponder}} L_{\text{ponder}}, \tag{7}$$

$$\lambda_{\text{conv}} = 0.5,\ \lambda_{\text{nt}} = 0.1,\ \lambda_{\text{div}} = 0.2,\ \lambda_{\text{dom}} = 0.1,\ \lambda_{\text{ponder}} = 0.01. \tag{8}$$

L_{task} is label-smoothed cross-entropy ($\varepsilon = 0.05$). L_{nt} encourages non-trivial first-round change: $\mathrm{ReLU}(\text{margin} - \|\mathbf{b}^{(1)} - \mathbf{b}^{(0)}\|)$, margin 0.35. L_{conv} penalises differences in later rounds: $\frac{1}{R-2}\sum_{t\geq 2}\|\mathbf{b}^{(t)} - \mathbf{b}^{(t-1)}\|$. L_{div} is pairwise cosine similarity among initial beliefs. L_{dom} penalises dominance in consensus weights: $\|\mathbb{E}[w_{\text{final}}] - 1/N\|^2$. L_{ponder} encourages early termination (Adaptive Computation Time).

5 Experiments

5.1 Datasets and Metrics

We primarily use SST-2 (binary sentiment, 67k/872 train/val). Additional results are reported on AG News (4-class, 120k/7.6k) and Amazon Reviews (5-class, 20k). Metrics: accuracy, Expected Calibration Error (ECE, 15 bins), robustness (accuracy after replacing 30% of tokens with random noise).

5.2 Model Configurations

We compare three settings:

- **BERT-tiny:** pre-trained encoder (4.4M) + linear classifier.
- **CIDA-BERT:** BERT-tiny + projection to $d = 64$ + CDP (5.3M).
- **CIDA V8 Tiny:** simple token embedder (vocab 8k, $d = 64$) + CDP, no pre-training (1.25M).

Training details: AdamW, LR=2e-5 (BERT) or 3e-4 (embedder), batch size 64/256, 30 epochs, early stopping (patience 8). All experiments run on a single RTX 3050 laptop GPU.

5.3 Main Results

Table 1: Comparison on SST-2 validation. ECE improvement relative to BERT-tiny.

Model	Params	Accuracy (%)	ECE (%)	Robustness (%)
BERT-tiny	4.4M	83.60	9.09	70.99
CIDA-BERT	5.3M	83.14	5.63 (↓38%)	68.58
CIDA V8 Tiny (no pre-train)	1.25M	76.66	3.41 (↓62%)	70.13

CIDA-BERT reduces ECE by 38% with negligible accuracy loss (Δ = 0.46%). The version trained from scratch on a simple embedder achieves the lowest ECE (3.41%) despite 4× fewer parameters than BERT-tiny, demonstrating that CDP intrinsically promotes calibration.

5.4 Ablation Study

To prove that CDP's effectiveness is due to its architecture and not merely extra parameters, we perform an ablation on a **simple token embedder** (no pre-training). The baseline embedder + linear classifier (1.04M) achieves only 50.9% accuracy (random guessing). Adding a CDP module (1.25M, +20% parameters) boosts accuracy to 76.7% and reduces ECE to 3.4% (Table **??**). Simply increasing the baseline's width to 1.25M yields no improvement, confirming the architectural nature of the gain.

5.4.1 Component Contribution

We disable individual CDP components and report averages over two random seeds (42, 123). Results are summarised in Table 2. Removing *Sparse Token Communication* hardly changes accuracy but worsens ECE by 30% (4.42% vs 3.41%), indicating that argument exchange is crucial for calibration. Disabling diversity loss ($\lambda_{\text{div}} = 0$) or dominance loss ($\lambda_{\text{dom}} = 0$) causes moderate ECE increases (5–8%). The optimal number of agents on SST-2 is $N = 2$, which yields the best ECE (3.40%) and robustness (70.9%).

Table 2: Ablation study on the simple embedder (mean over seeds 42, 123).

Variant	Params (M)	Accuracy (%)	ECE (%)	Robustness (%)
Baseline (no CDP)	1.04	50.92	5.69	50.92
CDP full	1.25	76.66	3.41	70.13
No communication	1.23	77.01	4.42	68.52
No diversity loss	1.25	77.41	3.58	70.41
No dominance loss	1.25	76.49	3.68	70.58
Agents=2	1.23	77.41	3.40	70.93
Agents=8	1.26	77.75	3.96	70.76

6 Limitations

- The main experiments use only 2 random seeds; for conclusive publication we will run 5 seeds and report confidence intervals.
- The comparison between BERT-tiny and CIDA-BERT is not strictly parameter-matched; a BERT-tiny + MLP baseline of equal size is required to isolate the effect of CDP.
- Results are shown primarily on SST-2 (binary classification). Generalisation to more complex NLU tasks (MNLI, QQP) remains to be verified.
- Our ablation uses a simple token embedder; the contribution of CDP components may differ when combined with a pre-trained encoder.

7 Conclusion

We presented CIDA, a novel architecture that embeds a collective deliberation protocol (CDP) into a text classifier. CDP lets multiple agents exchange theses, update their beliefs recurrently, and form a consensus weighted by their uncertainty. Experiments demonstrate that CDP substantially improves calibration: when added to BERT-tiny, ECE drops by 38% with almost no loss in accuracy. Moreover, an ablation on a simple embedder proves that CDP is not a mere parameter increase — it turns a random-guessing baseline into a well-performing model. We also isolated the role of each component: sparse communication is key for calibration, while diversity and dominance losses provide regularisation. The proposed architecture is reproducible and opens a new direction for building inherently calibrated neural networks.

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